

Alibek Kaliyev

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EXPERIENCE

Amazon Web Services

New York City, NY, USA

Software Development Engineer (Amazon Quick Suite Team)

January 2025 - Present

Java TypeScript AWS CDK AWS Fargate A2A Protocol Bedrock

- Contributed as a founding engineer for the launch of Amazon Quick Suite, a generative AI platform serving 500K+ monthly users; architected end-to-end conversational AI infrastructure including networking, APIs, automated testing frameworks (adopted by 3+ Quick Suite Teams), CI/CD pipelines, and performance benchmarking systems
- Drove zero-to-production deployment of Muse artifact generation agent in 1 week by conducting architectural trade-off analysis (A2A vs MCP protocols), partnering with Applied Science team to productionalize research models, and delivering executive demo that secured continued investment
- Pioneered multi-agent orchestration infrastructure by designing A2A server on internal Amazon framework, establishing reusable deployment patterns enabling standardized agent-to-agent communication at scale
- Built production infrastructure from ground up including VPC networking, DNS architecture, load balancers, security frameworks, and ECS Fargate orchestration; created cross-team testing framework that accelerated development velocity by 70% across Quick Suite organization
- Eliminated 6-8 hours/week of manual operational work through automated monitoring dashboards; led team in code quality with 584 reviews (41% of team total) averaging 1-hour turnaround, directly enabling rapid iteration cycles

Software Development Engineer (Amazon Q Business Team)

June 2024 - January 2025

Java TypeScript AWS CDK DynamoDB Lambda Kinesis

- Architected enterprise-grade active user metrics system (DAU, WAU, MTD/MAU) for Analytics Dashboard serving 1M+ customers across 4 regions with complex time-series aggregation, achieving 100% uptime over 1 year in production
- Identified and resolved critical data pipeline failure affecting 67 enterprise customers and 46K+ requests by deep-diving Kinesis error handling; coordinated cross-team remediation and mentored intern driving permanent solution
- Delivered 50% performance improvement for MS Teams Connector (4→6 files/sec) by architecting intelligent entity caching and preloading strategy, eliminating API bottlenecks during high-throughput crawling operations

SteppeTech Consulting

Remote

Co-founder & Technical Lead

January 2024 - March 2024

Python AWS React CLIP Elasticsearch

- Founded AI consulting firm and architected computer vision-powered trademark search platform for IP-Assist Patent Office using CLIP embeddings and AWS Elasticsearch, reducing search latency by 80% and achieving 65% cost reduction
- Designed cloud-native architecture integrating AWS services (S3, RDS, Lambda, VPC) with React frontend, enabling real-time similarity search across 100K+ trademark images while maintaining enterprise security standards

PROJECTS & RESEARCH

Capstone Project for Merck & Co.: Machine-Assisted Contextualization

Bethlehem, PA, USA

Python PyTorch Scikit-learn FastAPI AWS

January 2023 - December 2023

- Led ML pipeline development for pharmaceutical label classification achieving \$500K annual cost savings; improved prediction accuracy from 47% to 92% through advanced feature engineering, increasing labeling efficiency 10x
- Advanced semi-supervised GAN-BERT model to 96% accuracy; deployed production system on AWS with FastAPI supporting continuous model retraining and A/B testing capabilities

PUBLICATIONS

- A.T. Kaliyev**, R. Forelli, P. Sales, S. Qin, Y. Guo, S.O. Memik, M.W. Mahoney, A. Gholami, R.K. Vasudevan, S. Jesse, N. Tran, P. Harris, M. Takáč, J.C. Agar. Rapid Fitting of Band-Excitation Piezoresponse Force Microscopy Using Physics Constrained Unsupervised Neural Networks. *NeurIPS 2023 AI4Mat Workshop*.
- S. Qin, Y. Guo, **A.T. Kaliyev**, J.C. Agar. Why it is Unfortunate that Linear Machine Learning Models "Work" so well in Electromechanical Switching of Ferroelectric Thin Films. *Advanced Materials*. 2202814.

EDUCATION

University of Texas at Austin (GPA: 4.0/4.0)

Austin, TX, USA

Master of Science, Computer Science/ML - Earning degree while working full time

Expected December 2027

Lehigh University (GPA: 3.78/4.00, Magna Cum Laude)

Bethlehem, PA, USA

Bachelor of Science, Computer Science & Business (Minor: Cognitive Science)

August 2020 - May 2024